Stores & calculations

Follow the energy

1 Describe a change

Thrown ball: kinetic → gravitational; air resistance also warms surroundings.

Brakes: kinetic → thermal. Kettle: electrical transfer → water's thermal store.

In collisions, kinetic energy may decrease while total energy is conserved.

2 Conservation

Closed system: no net change in total energy. Include surroundings receiving transfers.

Dissipation spreads energy into less useful stores. "Wasted" does not mean destroyed.

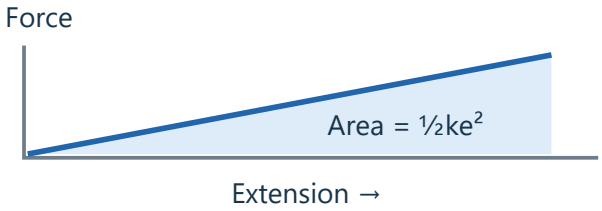
Use equations

3 Motion & height

$E_k = \frac{1}{2}mv^2$; $E_p = mg\Delta h$. Use given g. A 2 kg object at 3 m/s has 9 J.

Rearrange: $v = \sqrt{(2E_k/m)}$. Double speed → four times kinetic energy.

4 Springs



$E_e = \frac{1}{2}ke^2$ within proportional limit. e = stretched length − original length, in metres.

k = 200 N/m, e = 0.050 m → E_e = 0.25 J. Square the converted extension.

Reduce waste

5 Power is a rate

$P = E/t = W/t$; $E = Pt$. 1,800 J in 30 s → 60 W.

1 kW = 1,000 W; 1 kJ = 1,000 J. Powerful does not automatically mean efficient.

6 Efficiency

Useful output / total input, for energy or power. No unit; at most 1 or 100%.

350 J useful from 500 J → 70%; 150 J dissipated. Lubrication reduces friction and unwanted heating.